ForecastLabAI — forecasting model families

Problem

Name the implemented models, the shared interface, and which ones are optional — correctly.

Instant solution

All models implement `BaseForecaster` (`app/features/forecasting/models.py`): `fit()`, `predict()`, `get_params()`, `set_params()`.

Model	Class	Idea	Why it is there
Naive	`NaiveForecaster`	repeat last value	simplest baseline
Seasonal naive	`SeasonalNaiveForecaster`	repeat last season	recurring patterns
Moving average	`MovingAverageForecaster`	mean of recent values	noise reduction
Weighted MA	`WeightedMovingAverageForecaster`	recent values weigh more	recency
Seasonal average	`SeasonalAverageForecaster`	mean per seasonal position	seasonal baseline
Trend regression	`TrendRegressionBaselineForecaster`	fit a time trend	trend extrapolation
Regression (`regression`)	`RegressionForecaster`	`HistGradientBoostingRegressor` on features	tabular ML, exogenous what-if
Random forest	`RandomForestForecaster`	independent trees, averaged	nonlinear interactions
LightGBM	`LightGBMForecaster`	fast gradient-boosted trees	larger workloads
XGBoost	`XGBoostForecaster`	regularized boosted trees	advanced regression
Prophet-like	`ProphetLikeForecaster`	SimpleImputer + Ridge additive model	interpretable trend/season parts

Why it works / gotchas

- **Feature-flagged, off by default:** only `random_forest`, `lightgbm`, `xgboost` (`app/core/config.py`). LightGBM/XGBoost also need the extras `ml-lightgbm` / `ml-xgboost`.

- `prophet_like` is **custom scikit-learn**, not the Meta Prophet package, and it is not flagged.
- The "gradient-boosted regression" from the review is `RegressionForecaster` (model type `regression`).
- Baselines first: if a complex model cannot beat seasonal naive in backtests, its cost is not justified.
- Random forest = parallel independent trees (variance reduction); boosting = sequential trees fixing previous residuals.

Diagram brief

- Type: class/tree diagram.
- Root `BaseForecaster` (fit · predict · get_params · set_params).
- Three branches: Baselines (naive, seasonal naive, MA, weighted MA, seasonal average, trend regression) · ML (regression/HistGB, random forest, LightGBM, XGBoost) · Additive (prophet_like = Imputer + Ridge).
- Badge "flag, default off" on random forest, LightGBM, XGBoost; badge "extra" on LightGBM, XGBoost.